

Linear algebra, Math 1B03

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Short-term outline

- Review linear equations, section 1.1
- Introduce matrix notation, elimination method, section 1.2
- Introduce Matlab

Linear equations

- A linear equation in n unknowns or variables has the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where the x_i 's are the unknowns and the a_i 's and the b are numbers.

- We may use letters like x , y or z for the variables when we have only a few.
- If when you substitute $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ into the linear equation above and the left hand side equals the right hand side, we say that this is a solution to that linear equation. We call $(s_1, \dots, s_n)$ an ordered n -tuple.

$$\underline{2x_1 + 3x_2 = 4} \quad \leftarrow \text{linear equation}$$

$$x_1^2 + x_2^2 = 1 \quad \leftarrow \text{not linear.}$$

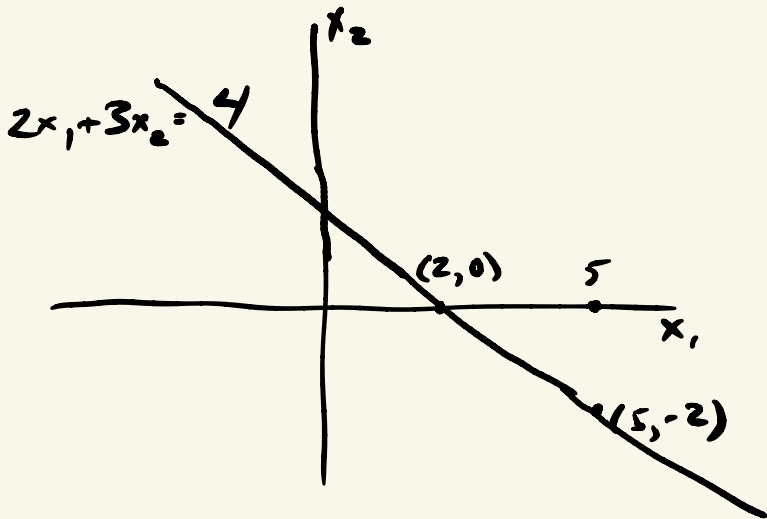
$x_1 = 5, x_2 = -2$ is a solution of

$$2x_1 + 3x_2 = 4.$$

$x_1 = 2, x_2 = 0$ is also a solution

Any point which lies on the line

$$2x_1 + 3x_2 = 4 \quad \text{is a solution}$$



Systems of linear equations

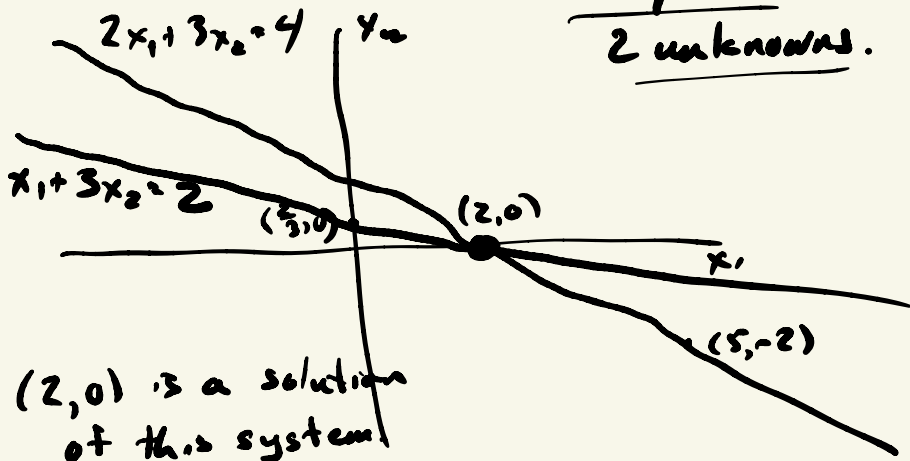
A finite set of linear equations is called a system of linear equations; in general a linear system looks like this:

$$\begin{array}{ccccccc} a_{11}x_1 + a_{12}x_2 + & \dots & + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + & \dots & + a_{2n}x_n & = & b_2 \\ & \vdots & & & \\ & & \ddots & & \\ a_{m1}x_1 + a_{m2}x_2 + & \dots & + a_{mn}x_n & = & b_m \end{array}$$

A solution to a linear system is an ordered n -tuple which is simultaneously a solution to each equation.

Ex:
$$\begin{cases} 2x_1 + 3x_2 = 4 \\ x_1 + 3x_2 = 2 \end{cases} \text{ a system of linear equations}$$

2 equations,
2 unknowns.



$(2, 0)$ is a solution
of this system.

Ex

$$\textcircled{1} \quad \left. \begin{array}{l} 2x_1 + 3x_2 = 4 \\ 2x_1 + 3x_2 = 5 \end{array} \right\} \begin{array}{l} \text{system of l.n. eqs.} \\ \text{parallel lines which} \\ \text{are not equal.} \end{array}$$

$$\textcircled{2} \quad \left. \begin{array}{l} 2x_1 + 3x_2 = 4 \\ \hline 4x_1 + 6x_2 = 8 \end{array} \right\} \begin{array}{l} \text{same line so} \\ \text{there are infinitely} \\ \text{many solutions} \end{array}$$

Augmented matrices

For the linear system

$$\begin{array}{cccccc} a_{11}x_1 + a_{12}x_2 + & \dots & + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + & \dots & + a_{2n}x_n & = & b_2 \\ \vdots & \ddots & \vdots & & \\ a_{m1}x_1 + a_{m2}x_2 + & \dots & + a_{mn}x_n & = & b_m \end{array}$$

the augmented matrix is

$$\left(\begin{array}{ccccc} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \ddots & \vdots & & \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right)$$

Without the right most column, we call this the coefficient matrix. It has m rows and n columns and we say it is $m \times n$.